TrustHire: Real-Time Multimodal Detection of Synthetic and Impersonated Candidates in Remote Job Interviews

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***Abstract*—Remote hiring now has to deal with candidates who are not who they claim to be: synthetic talking heads, cloned voices, stand-in proxies, and applicants reading prepared answers off a second screen. Detectors built for one of these problems are usually evaluated on studio-quality benchmarks that look little like a webcam interview. We present TrustHire, a system that watches a live interview over WebRTC, cuts it into 3 s windows with 50% overlap, and scores each window with ten evidence modules covering face forgery, voice spoofing, lip synchrony, face and voice identity continuity, gaze behaviour, remote photoplethysmography, scene objects, answer delivery, and consistency with the resume. A calibrated fusion stage turns these into a per-window risk timeline and a session trust score, and every alert is backed by a visual or acoustic explanation. On public data, the selected voice-spoofing model reaches 1.25% equal error rate (EER) on In-the-Wild against 42.50% for the commonly used AASIST baseline, while degrading sharply in reverberant rooms. On a pilot set of 18 interview recordings collected by the authors, replacing a FaceForensics++-trained detector with a general AI-image detector and learning the fusion weights raises session-level AUC from 0.714 to 0.938 $\pm$ 0.019. The full pipeline runs at 1.12 s per window on a laptop CPU, faster than the 1.5 s hop. We report where the system fails, most notably on scripted reading.**

***Index Terms*—deepfake detection, voice anti-spoofing, multimodal fusion, remote hiring, audio-visual synchrony, identity verification**

I. Introduction

Video interviews have become the default first round of hiring, and the trust that used to come from sitting in the same room has not been replaced by anything comparable. In 2022 the FBI warned that applicants were using deepfake video and stolen identities to win remote technical roles [1]. Since then, text-to-video avatar services and few-shot voice cloning have made convincing fakes cheaper still. Fraud in interviews is also not only a deepfake problem: a different person can take the call after a genuine applicant has been verified, or a real candidate can read answers written by someone else.

Most published detectors address a single modality and are tested on curated datasets such as FaceForensics++ [2] or ASVspoof [3]. A recurring finding is that their accuracy drops once the test data comes from somewhere else [4], [5]. An interviewer, meanwhile, needs something different from a benchmark score: a running indication of *when* something looks wrong, *which* signal raised the concern, and how sure the system is.

This paper describes TrustHire, a working system built around that need, and the measurements we made while building it. Our contributions are:

- a streaming architecture that analyses live WebRTC video and audio in overlapping 3 s windows with ten evidence modules, each reporting a score *and* a confidence, so missing or unreliable evidence (no face, no speech, heavy noise) is handled explicitly rather than imputed;
- a component study on public data, including a robustness analysis showing that a strong speech anti-spoofing model fails under room reverberation and that neither front-end dereverberation nor augmentation-based fine-tuning fixed it;
- an interview-shaped pilot evaluation in which the commonly used face-forgery model turned out to be anti-correlated with the ground truth, a general AI-image detector separated real and generated videos, and a learned, calibrated fusion improved session-level detection; we also show how isotonic calibration on small folds silently destroyed ranking performance;
- engineering measures that bring the full pipeline below real time on a CPU-only laptop.

II. Related Work

Face forgery detection. Frame-level CNNs such as XceptionNet [6] and EfficientNet [7] trained on FaceForensics++ [2] remain standard baselines. DeepfakeBench [5] compared many detectors under a common protocol and reported large drops on unseen datasets such as Celeb-DF [8] and DFDC [9]. Detectors trained on many generative models, for example linear probes over CLIP features [10], generalise better to fully synthetic images, which is closer to the avatars produced by current text-to-video tools.

Voice anti-spoofing. AASIST [11] and RawNet2 [12] operate on raw waveforms and perform very well on ASVspoof 2019 LA [3]. Müller et al. [4] showed that such models break down on In-the-Wild recordings of public figures. Front-ends based on self-supervised speech models such as XLS-R [13] narrow this gap; the Spectra-0 model [14] we use combines XLS-R with an ECAPA-TDNN head.

Audio-visual consistency. SyncNet [15] learns a joint embedding of mouth crops and MFCC frames and estimates the audio-visual offset; AV-HuBERT [16] learns stronger joint representations at a much higher compute cost. Multimodal deepfake datasets such as FakeAVCeleb [17] and LAV-DF [18] combine visual and audio manipulations.

Identity and liveness. ArcFace [19] with SCRFD detection [20] and ECAPA-TDNN [21] are widely used for face and speaker verification. Remote photoplethysmography methods such as CHROM [22] and POS [23] recover a pulse signal from skin colour and have been used as liveness cues.

To our knowledge, prior work rarely combines these signals in a single streaming system aimed at interviews or evaluates them on interview-like recordings, which is the gap this work targets. Multimodal fusion in general is surveyed in [24].

III. System Design

A. Streaming pipeline

The browser sends camera and microphone streams over WebRTC to a FastAPI server, where an aiortc receiver resamples video onto a constant 25 fps grid (needed by SyncNet) and audio to 16 kHz mono (Fig. 1). Video frames are spaced using the sender's RTP clock rather than their arrival time. We found this mattered: with arrival-time stamps, bursty delivery under CPU load scrambled the lip motion and SyncNet's confidence fell from about 8 to below 2 on a clip that was perfectly synchronised offline. A chunker emits 3 s windows every 1.5 s. Results are pushed to a React dashboard over a WebSocket; recorded interviews follow the same path through an offline job queue.

B. Evidence modules

Table I lists the modules. All of them report a risk $s_m \in [0, 1]$, where higher means more suspicious, together with a confidence $c_m \in [0, 1]$ that reflects the quality of the evidence: face size and coverage for the video branches, the fraction of speech found by voice activity detection, and an estimated speech-to-noise ratio for the anti-spoofing branch. A module with nothing to measure returns $c_m = 0$ and is treated as missing.

TABLE I
Evidence modules and the models behind them

Module	Model	What it flags
Video forgery	SwinV2-L AI-image classifier [25], [26]	generated / swapped faces
Voice spoofing	Spectra-0 (XLS-R + ECAPA) [14]	synthetic or converted speech
Lip synchrony	SyncNet [15]	dubbed or misaligned audio
Face identity	SCRFD + ArcFace R50 [19], [20]	person changed after enrolment
Voice identity	ECAPA-TDNN [21]	speaker changed
Behaviour	MediaPipe Face Mesh [27]	off-screen gaze, reading sweeps
Liveness	rPPG, POS + CHROM [22], [23]	absent pulse signal
Scene objects	YOLO26n (COCO)	phone, second person, extra screen
Answer delivery	Whisper [28] + prosody	fluency typical of reading
Resume check	embedding retrieval + NLI	contradictions with the CV

Identity continuity compares each window both with the enrolled reference and with a running average of the session, so it can detect a person who was wrong from the start as well as one who changes mid-interview. Lip synchrony is scored against the session's own median offset, because genuine setups often carry a constant skew (Bluetooth headsets add 150–250 ms); offsets larger than 320 ms remain suspicious regardless.

C. Fusion and calibration

For each window t the fusion stage receives $\{(s_m, c_m)\}$ for all modules plus a few raw measurements (sync confidence, identity similarities, speech SNR). We compared three fusers. The first is a confidence-weighted mean of log-odds with an additional term that lets a single strong module dominate:

$$z_t = b + \frac{\sum_m w_m c_m \ell_m}{\sum_m w_m c_m} + \sum_m v_m c_m \max(0, \ell_m), \quad (1)$$

where $\ell_m = \mathrm{logit}(s_m)$ and the weights $w_m, v_m \geq 0$ are fitted by logistic loss. The mean term stops correlated modules that all report "genuine" from compounding into unrealistic certainty, a problem we saw with a plain naive-Bayes sum. The second fuser is a LightGBM stacker [29] over scores, confidences, raw measurements and rolling statistics of the previous five windows, which handles missing modules natively. The third is a small transformer that treats each module in the current and three previous windows as a token and is trained with random modality dropout. Outputs are calibrated with temperature scaling [30] fitted on held-out participants. The session trust score is $100\,(1 - R)$ with $R = 0.6\,\overline{r}_{\mathrm{top10\%}} + 0.4\,\overline{r}$, so a short burst of strong evidence matters while one noisy window does not decide the outcome.

D. Explanations

Every alert links to its evidence. For the video branch we compute Grad-CAM [31] on the classifier's third stage

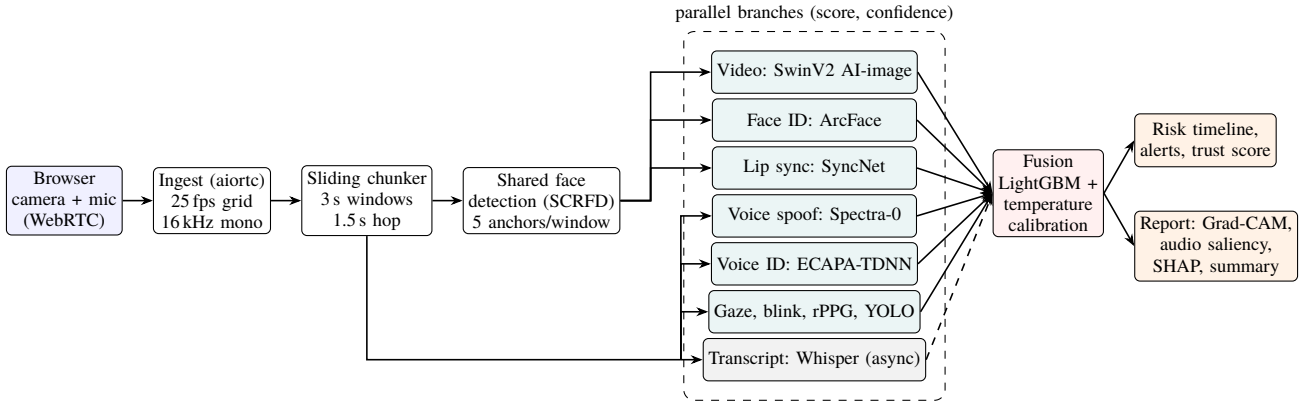


Fig. 1. TrustHire architecture. Each window is analysed by parallel branches that share face detections; each branch returns a score in $[0, 1]$ and a confidence. The transcript branch runs asynchronously and updates earlier windows when an answer has been transcribed.



Fig. 2. Grad-CAM for the video branch on two AI-generated avatars from the pilot set (original and overlay). Only synthetic faces are shown.

TABLE II
VOICE ANTI-SPOOFING ON PUBLIC DATA (EER %, AUC IN BRACKETS)

Model	In-the-Wild (n=400)	ASVspoof 2019 LA (n=308)
AASIST [11]	42.50 (0.580)	1.62 (0.998)
AASIST-L [11]	43.50 (0.618)	2.27 (0.999)
Spectra-0 [14]	1.25 (0.998)	0.00 (1.000)
Spectra-0, int8 (deployed)	1.25 (0.998)	0.00 (1.000)

(Fig. 2); for audio, an occlusion map shows which quarter-second slices raised the spoof probability; for the LightGBM fuser, SHAP values [32] give each module's contribution. A short natural-language summary is generated from these numbers and exported with the timeline as a PDF report. The report states that it is evidence for a reviewer, not a decision.

E. Running below real time on a CPU

The deployment target was a laptop without a GPU (Intel Core i5-1235U). A first version needed 9.5 s per window. Three changes made the difference (Fig. 9). Flushing denormal floating-point values to zero in ONNX Runtime made the convolutional models up to forty times faster. Capping each concurrent branch at three threads removed oversubscription. Finally, because consecutive windows overlap by half, running the most expensive modules on alternate windows still covers every instant of the interview while halving their cost; in between, the last result is carried forward. The SwinV2 and Spectra-0 models are additionally quantised to int8, which cut their latency by factors of 1.9 and 1.6 without changing their accuracy on our evaluation sets.

IV. Experimental Setup

Public data. Voice anti-spoofing was evaluated on stratified subsets of In-the-Wild [4] (400 utterances, two shards) and the ASVspoof 2019 LA evaluation set [3] (308 utterances). Each utterance was scored on its first 4.04 s, the same context the live system uses. Robustness conditions were applied to the same utterances: additive white and pink noise, Opus at 16 kb/s (typical of WebRTC), MP3 at 32 kb/s, telephone band-limiting, and synthetic room reverberation with RT_{60} of 0.25–0.7 s. Face verification used the 1000 LFW test pairs [33], run end-to-end through detection and alignment; speaker verification used 2000 trials of 4 s segments from LibriSpeech test-clean [34].

Pilot interview set. The authors recorded 18 videos (339 analysis windows): five genuine webcam answers and two re-encoded copies of them (480p and 1440p), three answers read from a prepared script, two clips with replaced or misaligned audio, and six fully AI-generated talking heads from a text-to-video service. All real footage shows one consenting participant. The re-encoded copies were placed in the same cross-validation group as their originals so that no model is tested on a copy of a video it was trained on.

Protocol and metrics. We report AUC and EER, recall at a fixed false-positive rate (FPR), and latency. Fusion models were evaluated with grouped five-fold cross-validation; inside each training fold, a fifth of the groups were held out to fit the calibrator. Because the pilot set is small, every fusion result is repeated over five random seeds and reported as mean $\pm$ standard deviation.

V. Results

A. Voice anti-spoofing and robustness

Table II confirms the generalisation gap reported in [4]. AASIST is nearly perfect on ASVspoof but close to chance on In-the-Wild, whereas Spectra-0 stays near 1% EER on both. Dynamic int8 quantisation of Spectra-0 left the EER unchanged and made it 1.7 times faster on our CPU, so the quantised model is deployed.

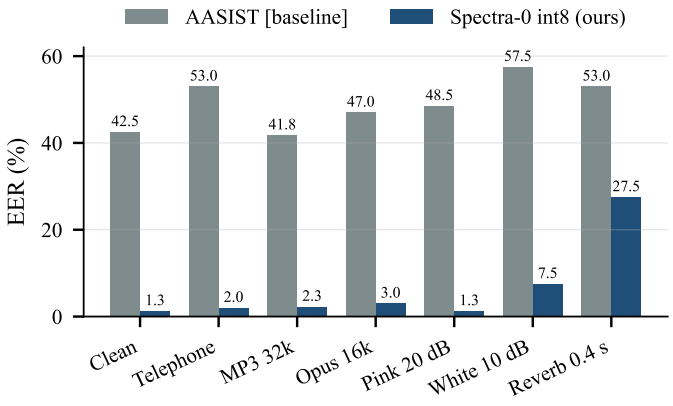


Fig. 3. EER on the same In-the-Wild utterances under channel degradations.

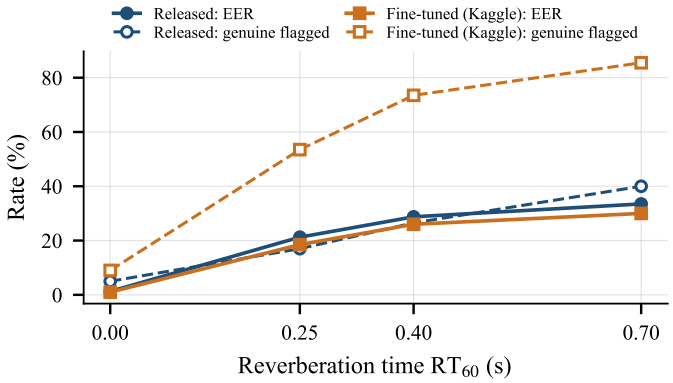


Fig. 4. Effect of room reverberation on the voice-spoofing module before and after augmentation-based fine-tuning. Solid lines: EER. Dashed lines: share of genuine speakers with spoof probability above 0.5.

Codecs cost little and noise a moderate amount (Fig. 3): Opus raised the EER to 3.00%, and white noise at 10 dB to 7.50%. Reverberation was a different story. At $RT_{60} = 0.4$ s, typical of an ordinary furnished room, the EER rose to 27.50% and more than a quarter of genuine speakers crossed the 0.5 decision threshold (Fig. 4). Two remedies failed. WPE dereverberation [35] in front of the detector gave 26.50% EER. Fine-tuning Spectra-0 for two epochs on ASVspoof 2019 LA with reverberation, noise and codec augmentation (on a Tesla T4) lowered the EER slightly but shifted the score distribution, so that 54–86% of reverberant genuine speakers were flagged. We kept the released model and instead lower this module's confidence when the estimated speech-to-noise ratio is poor.

B. Identity and lip synchrony

On LFW the end-to-end face path reached an AUC of 0.986 and 3.0% EER (accuracy 96.9%). This is below the accuracy usually quoted for ArcFace, which is measured on pre-aligned crops; at the deployed similarity threshold no impostor pair was accepted, and 3.6% of same-person pairs would be flagged. Speaker verification on LibriSpeech gave an AUC of 0.998 and 1.8% EER. For lip synchrony we shifted the audio of a reference talking-head clip by -400 to $+400$ ms; the estimated offset was within one frame of the truth in every window, and the sync risk rose monotonically with the shift (Fig. 5). This check uses a single clip and should be read as a sanity test, not a benchmark.

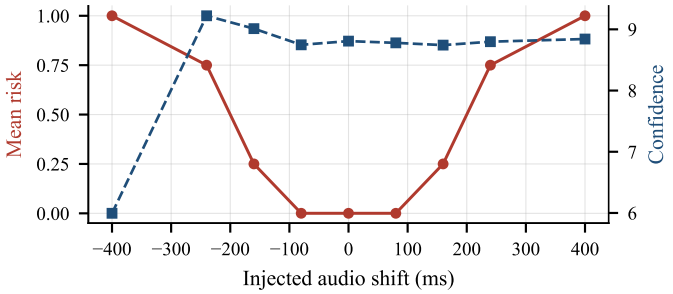


Fig. 5. Lip-synchrony risk and SyncNet confidence versus an injected audio shift.

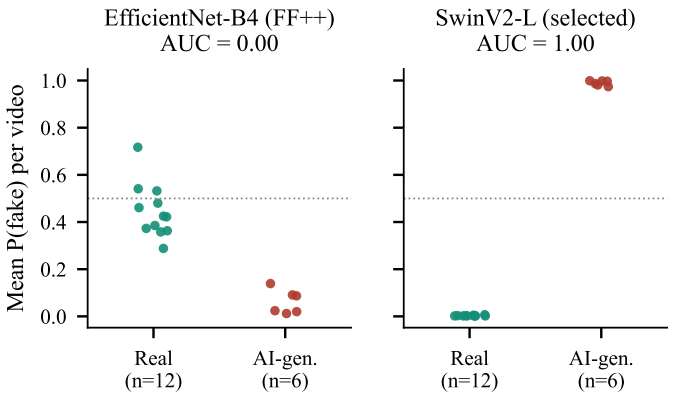


Fig. 6. Mean per-video fake probability on the pilot set for the FaceForensics++-trained baseline and the selected detector.

C. Pilot interview set

Video detector selection. The DeepfakeBench EfficientNet-B4 trained on FaceForensics++ was, on our recordings, anti-correlated with the truth: face crops from genuine webcam video received higher fake probabilities than the AI-generated avatars (Fig. 6, AUC 0.00). We benchmarked eight publicly available detectors on the same frames. A SwinV2-L classifier trained to separate AI-generated from real images [26] gave every real video a mean fake probability below 0.01 and every generated video one above 0.96. With six generated and twelve real videos this perfect separation is encouraging but not conclusive, and the set contains no real-time face swaps.

Fusion. With the new video detector and the hand-set fusion weights, genuine videos received trust scores of 95 to 99 and generated videos 0 to 2. Session-level AUC stayed low, however, because audio-mismatch and scripted clips were scored as genuine. One dubbed clip, for example, obtained a trust score of 97.6 although its lip-sync risk was 0.82, higher than that of any genuine video; the fixed weights simply gave lip synchrony too little influence. Learning the weights fixed most of this (Fig. 7). Table III summarises the comparison.

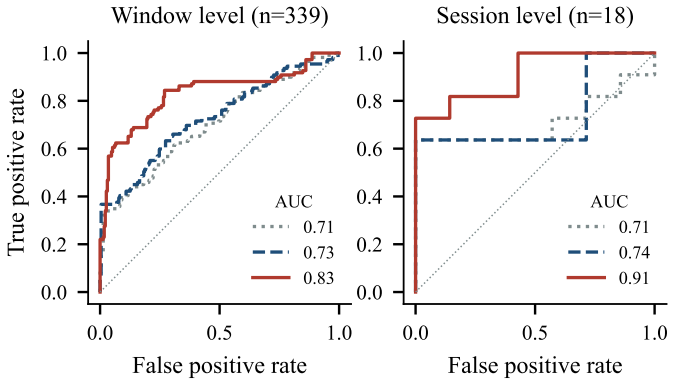


Fig. 7. Cross-validated ROC curves on the pilot set (learned fusion shown for one seed).

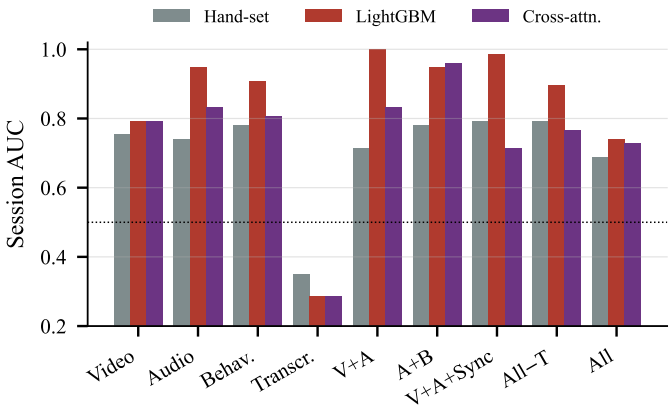


Fig. 8. Session-level AUC on the pilot set for modality subsets and fusers (grouped cross-validation).

TABLE III
PILOT-SET DETECTION, GROUPED CROSS-VALIDATION

Configuration	Window AUC	Session AUC	Sess. EER (%)
Hand-set, FF++	0.705	0.714	46.8
Hand-set, SwinV2	0.728	0.740	–
Learned (5 seeds)	0.835 ± 0.021	0.938 ± 0.019	16.1 ± 5.7
Learned fusion, recall at 5% window FPR (mean of 5 seeds)			
AI-generated	84.2%		
Audio mismatch	70.4%		
Scripted reading	10.0%		
Genuine (flagged)	3.9%		

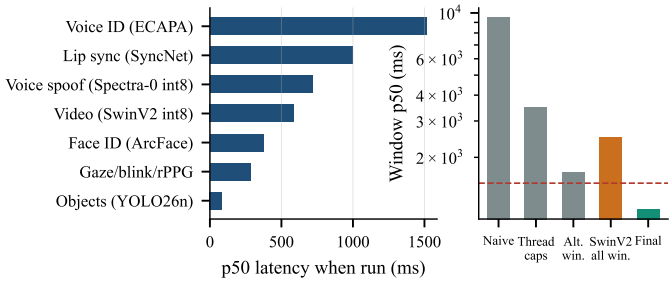


Fig. 9. Left: median module latency when the module runs. Right: window latency after each optimisation step (log scale); the dashed line is the 1.5 s hop.

Calibration matters for ranking. Our first learned fuser used isotonic regression [36], fitted on the three or four videos available for calibration in each fold. Isotonic regression is a step function, and on so few videos it mapped 339 window scores onto only 52 to 112 distinct values. The resulting ties lowered and destabilised the ranking: over five seeds, session AUC was 0.886 ± 0.083, against 0.938 ± 0.019 with rank-preserving temperature scaling. We now use temperature scaling for every fuser.

Ablation. Fig. 8 compares modality subsets. The transcript features hurt every combination they were part of; on their own they ranked sessions worse than chance. On spoken answers the AI-text detector assigned probabilities above 0.95 to several genuine responses, and the fluency features overlapped completely between scripted and spontaneous answers. The deployed fusion therefore excludes transcript features from the score but still shows them to the reviewer. No module separated scripted reading from genuine answers in this set.

D. Latency

With all modules enabled, the median processing time is 1.12 s per 3 s window (95th percentile 2.12 s), a real-time factor of 0.75, with a peak memory footprint of 2.4 GB (Fig. 9). The transcript branch is excluded from this figure because it runs asynchronously when an answer ends.

VI. Discussion and Limitations

The clearest lesson is that component benchmarks did not predict behaviour on our own recordings. The face-forgery model had published cross-dataset results, yet it ranked webcam faces as more fake than generated avatars. The voice model had excellent In-the-Wild numbers, yet it struggled with a reverberant room. Evaluating on data that looks like the deployment setting changed two of our main design decisions.

The pilot set is small and narrow: 18 videos, a single real participant, two rooms, one text-to-video generator, and no real-time face swaps. The cross-validated numbers are therefore indicative only, and some behavioural cues may partly encode the room rather than the attack. Scripted reading remains undetected. The pulse-based liveness signal is unreliable under compression and is only shown as low-severity information. Several components carry non-commercial or copyleft licences. Finally, accent-dependent features such as fillers and pause regularity risk unfair treatment of non-native speakers; this is one reason the score excludes transcript features.

We see the system as decision support. Alerts are explained, every session is meant to be reviewed by a person, and monitoring of this kind requires informed consent from candidates and compliance with biometric data regulations.

VII. Conclusion

TrustHire combines ten forms of evidence into a calibrated, explained risk timeline for live interviews and runs below real

time on a laptop CPU. Swapping the face-forgery model for a general AI-image detector and learning the fusion weights lifted session-level AUC on our pilot set from 0.714 to 0.938. The next steps are a larger, multi-participant recording set with real-time face swaps and more scripted sessions, and voice anti-spoofing training data that includes genuinely reverberant speech.

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References

[1] Federal Bureau of Investigation, Internet Crime Complaint Center, "Deepfakes and stolen PII utilized to apply for remote work positions," Public Service Announcement I-062822-PSA, Jun. 2022.

[2] A. Rössler, D. Cozzolino, L. Verdoliva, C. Riess, J. Thies, and M. Nießner, "FaceForensics++: Learning to detect manipulated facial images," in *Proc. IEEE/CVF Int. Conf. Computer Vision (ICCV)*, 2019, pp. 1–11.

[3] X. Wang, J. Yamagishi, M. Todisco *et al.*, "ASVspoof 2019: A large-scale public database of synthesized, converted and replayed speech," *Computer Speech & Language*, vol. 64, p. 101114, 2020.

[4] N. M. Müller, P. Czempin, F. Dieckmann, A. Froghyar, and K. Böttinger, "Does audio deepfake detection generalize?" in *Proc. Interspeech*, 2022, pp. 2783–2787.

[5] Z. Yan, Y. Zhang, X. Yuan, S. Lyu, and B. Wu, "DeepfakeBench: A comprehensive benchmark of deepfake detection," in *Advances in Neural Information Processing Systems (NeurIPS), Datasets and Benchmarks Track*, 2023.

[6] F. Chollet, "Xception: Deep learning with depthwise separable convolutions," in *Proc. IEEE Conf. Computer Vision and Pattern Recognition (CVPR)*, 2017, pp. 1251–1258.

[7] M. Tan and Q. V. Le, "EfficientNet: Rethinking model scaling for convolutional neural networks," in *Proc. Int. Conf. Machine Learning (ICML)*, 2019, pp. 6105–6114.

[8] Y. Li, X. Yang, P. Sun, H. Qi, and S. Lyu, "Celeb-DF: A large-scale challenging dataset for DeepFake forensics," in *Proc. IEEE/CVF Conf. Computer Vision and Pattern Recognition (CVPR)*, 2020, pp. 3207–3216.

[9] B. Dolhansky, J. Bitton, B. Pflaum, J. Lu, R. Howes, M. Wang, and C. C. Ferrer, "The DeepFake detection challenge (DFDC) dataset," *arXiv preprint arXiv:2006.07397*, 2020.

[10] U. Ojha, Y. Li, and Y. J. Lee, "Towards universal fake image detectors that generalize across generative models," in *Proc. IEEE/CVF Conf. Computer Vision and Pattern Recognition (CVPR)*, 2023, pp. 24 480–24 489.

[11] J.-w. Jung, H.-S. Heo, H. Tak, H.-j. Shim, J. S. Chung, B.-J. Lee, H.-J. Yu, and N. Evans, "AASIST: Audio anti-spoofing using integrated spectro-temporal graph attention networks," in *Proc. IEEE Int. Conf. Acoustics, Speech and Signal Processing (ICASSP)*, 2022, pp. 6367–6371.

[12] H. Tak, J. Patino, M. Todisco, A. Nautsch, N. Evans, and A. Larcher, "End-to-end anti-spoofing with RawNet2," in *Proc. IEEE Int. Conf. Acoustics, Speech and Signal Processing (ICASSP)*, 2021, pp. 6369–6373.

[13] A. Babu, C. Wang, A. Tjandra *et al.*, "XLS-R: Self-supervised cross-lingual speech representation learning at scale," in *Proc. Interspeech*, 2022, pp. 2278–2282.

[14] lab260, "Spectra-0: XLS-R and ECAPA-TDNN speech anti-spoofing model," Hugging Face model card, https://huggingface.co/lab260/spectra_0, 2025.

[15] J. S. Chung and A. Zisserman, "Out of time: Automated lip sync in the wild," in *Computer Vision – ACCV 2016 Workshops*, ser. Lecture Notes in Computer Science, vol. 10117, 2017, pp. 251–263.

[16] B. Shi, W.-N. Hsu, K. Lakhotia, and A. Mohamed, "Learning audio-visual speech representation by masked multimodal cluster prediction," in *Proc. Int. Conf. Learning Representations (ICLR)*, 2022.

[17] H. Khalid, S. Tariq, M. Kim, and S. S. Woo, "FakeAVCeleb: A novel audio-video multimodal deepfake dataset," in *Advances in Neural Information Processing Systems (NeurIPS), Datasets and Benchmarks Track*, 2021.

[18] Z. Cai, K. Stefanov, A. Dhall, and M. Hayat, "Do you really mean that? content driven audio-visual deepfake dataset and multimodal method for temporal forgery localization," in *Proc. Int. Conf. Digital Image Computing: Techniques and Applications (DICTA)*, 2022.

[19] J. Deng, J. Guo, N. Xue, and S. Zafeiriou, "ArcFace: Additive angular margin loss for deep face recognition," in *Proc. IEEE/CVF Conf. Computer Vision and Pattern Recognition (CVPR)*, 2019, pp. 4690–4699.

[20] J. Guo, J. Deng, A. Lattas, and S. Zafeiriou, "Sample and computation redistribution for efficient face detection," in *Proc. Int. Conf. Learning Representations (ICLR)*, 2022.

[21] B. Desplanques, J. Thienpondt, and K. Demuynck, "ECAPA-TDNN: Emphasized channel attention, propagation and aggregation in TDNN based speaker verification," in *Proc. Interspeech*, 2020, pp. 3830–3834.

[22] G. de Haan and V. Jeanne, "Robust pulse rate from chrominance-based rPPG," *IEEE Transactions on Biomedical Engineering*, vol. 60, no. 10, pp. 2878–2886, 2013.

[23] W. Wang, A. C. den Brinker, S. Stuijk, and G. de Haan, "Algorithmic principles of remote PPG," *IEEE Transactions on Biomedical Engineering*, vol. 64, no. 7, pp. 1479–1491, 2017.

[24] T. Baltrušaitis, C. Ahuja, and L.-P. Morency, "Multimodal machine learning: A survey and taxonomy," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 41, no. 2, pp. 423–443, 2019.

[25] Z. Liu, H. Hu, Y. Lin, Z. Yao, Z. Xie, Y. Wei, J. Ning, Y. Cao, Z. Zhang, L. Dong, F. Wei, and B. Guo, "Swin transformer V2: Scaling up capacity and resolution," in *Proc. IEEE/CVF Conf. Computer Vision and Pattern Recognition (CVPR)*, 2022, pp. 12 009–12 019.

[26] haywoodsloan, "AI image detector (SwinV2)," Hugging Face model card, https://huggingface.co/haywoodsloan/ai-image-detector-deploy, 2025.

[27] Y. Kartynnik, A. Ablavatski, I. Grishchenko, and M. Grundmann, "Real-time facial surface geometry from monocular video on mobile GPUs," in *CVPR Workshop on Computer Vision for Augmented and Virtual Reality*, 2019.

[28] A. Radford, J. W. Kim, T. Xu, G. Brockman, C. McLeavey, and I. Sutskever, "Robust speech recognition via large-scale weak supervision," in *Proc. Int. Conf. Machine Learning (ICML)*, 2023, pp. 28 492–28 518.

[29] G. Ke, Q. Meng, T. Finley, T. Wang, W. Chen, W. Ma, Q. Ye, and T.-Y. Liu, "LightGBM: A highly efficient gradient boosting decision tree," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2017, pp. 3146–3154.

[30] C. Guo, G. Pleiss, Y. Sun, and K. Q. Weinberger, "On calibration of modern neural networks," in *Proc. Int. Conf. Machine Learning (ICML)*, 2017, pp. 1321–1330.

[31] R. R. Selvaraju, M. Cogswell, A. Das, R. Vedantam, D. Parikh, and D. Batra, "Grad-CAM: Visual explanations from deep networks via gradient-based localization," in *Proc. IEEE Int. Conf. Computer Vision (ICCV)*, 2017, pp. 618–626.

[32] S. M. Lundberg and S.-I. Lee, "A unified approach to interpreting model predictions," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2017, pp. 4765–4774.

[33] G. B. Huang, M. Ramesh, T. Berg, and E. Learned-Miller, "Labeled faces in the wild: A database for studying face recognition in unconstrained environments," University of Massachusetts, Amherst, Tech. Rep. 07-49, 2007.

[34] V. Panayotov, G. Chen, D. Povey, and S. Khudanpur, "Librispeech: An ASR corpus based on public domain audio books," in *Proc. IEEE Int. Conf. Acoustics, Speech and Signal Processing (ICASSP)*, 2015, pp. 5206–5210.

[35] T. Nakatani, T. Yoshioka, K. Kinoshita, M. Miyoshi, and B.-H. Juang, "Speech dereverberation based on variance-normalized delayed linear prediction," *IEEE Transactions on Audio, Speech, and Language Processing*, vol. 18, no. 7, pp. 1717–1731, 2010.

[36] B. Zadrozny and C. Elkan, "Transforming classifier scores into accurate multiclass probability estimates," in *Proc. ACM SIGKDD Int. Conf. Knowledge Discovery and Data Mining*, 2002, pp. 694–699.